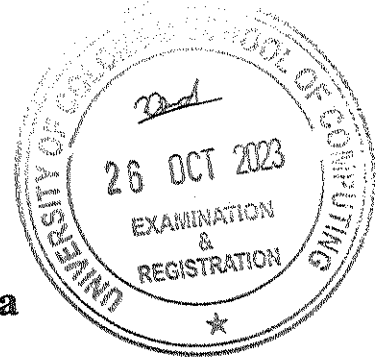




University of Colombo, Sri Lanka

University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Second Year Examination - Semester I - UCSC AY20 [held in October 2023]

SCS2203 – Software Engineering III

(Two (2) Hours)

Answer ALL questions

Number of Pages = 15

Number of Questions = 4

060



To be completed by the candidate

Index Number:

--	--	--	--	--	--	--	--

Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 15 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smartwatches, and mobile phones are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Index No:

Question 01 (20 Marks)

(a) State whether the following statements are true or false. [4 marks]

(i) Files as a Service (FaaS) is a category of Cloud Computing.

(ii) In IaaS, applications are hosted by a third party to offer to customers.

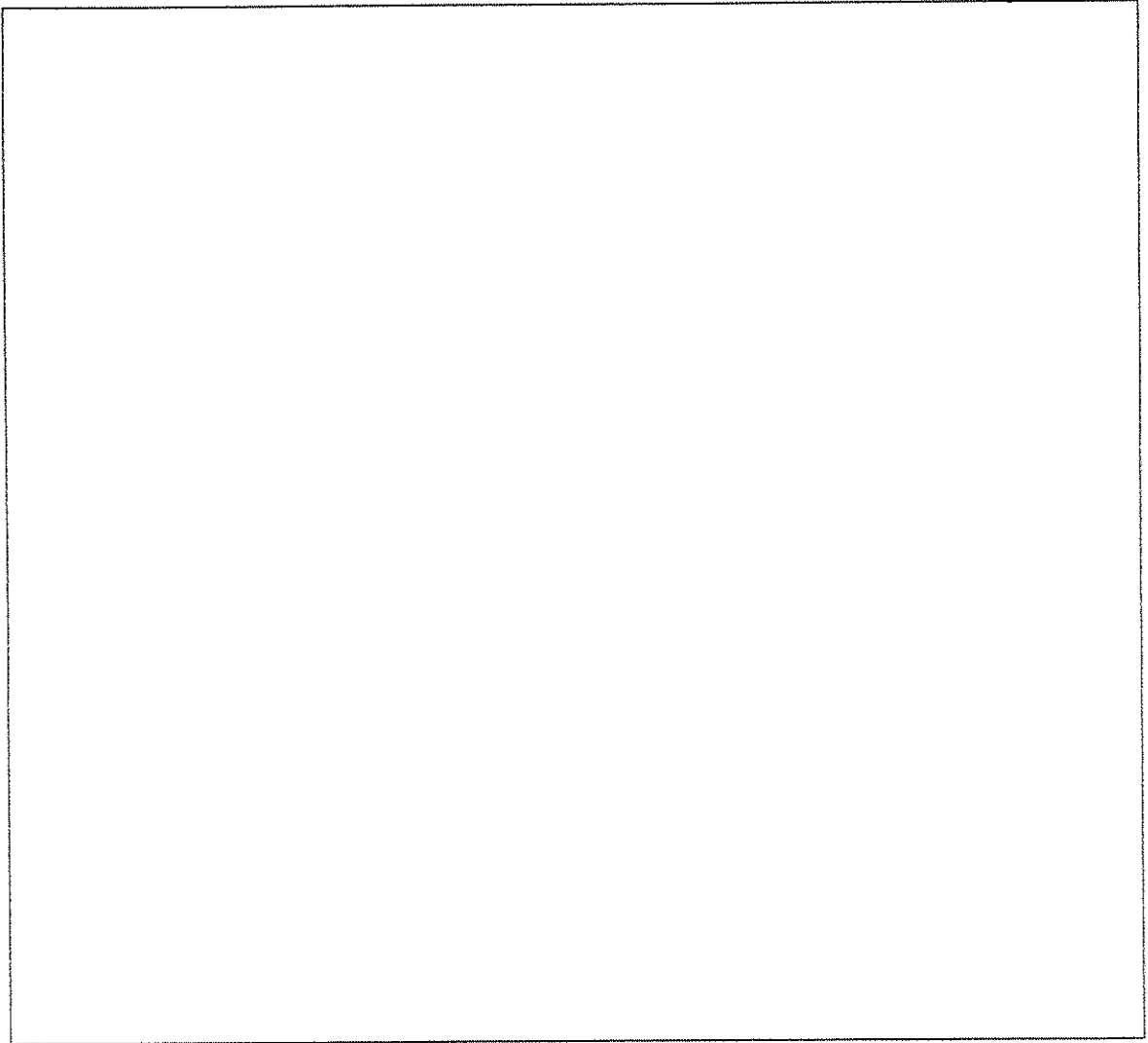
(iii) Hosted applications are an example of the Platform as a Service (PaaS).

(iv) With a VM, multiple copies of the VM images cannot be made, each with its own IP address and host ID.

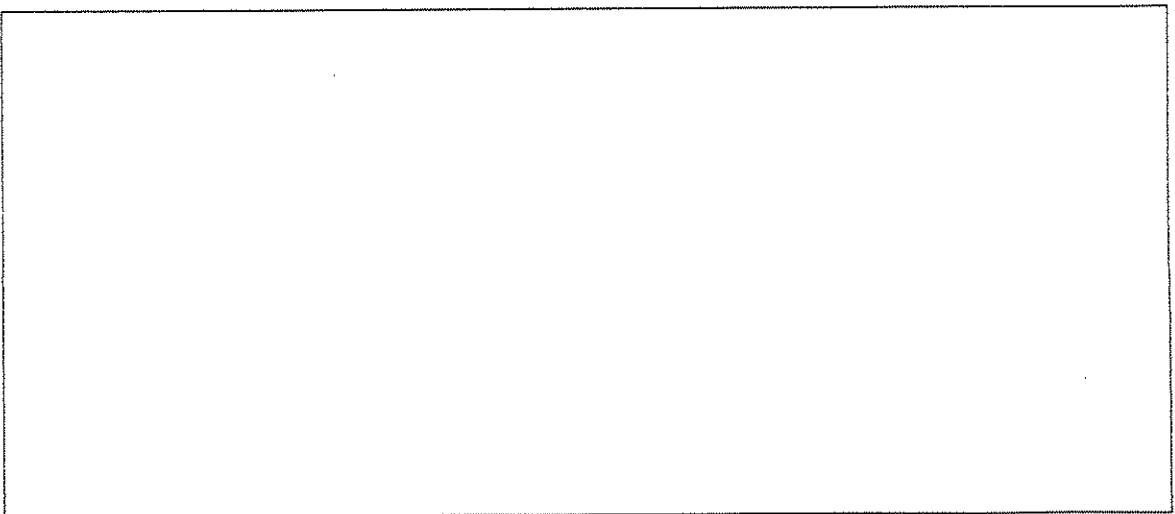
(b) What are Business Process Management Suites and briefly explain their categories?

[7 marks]

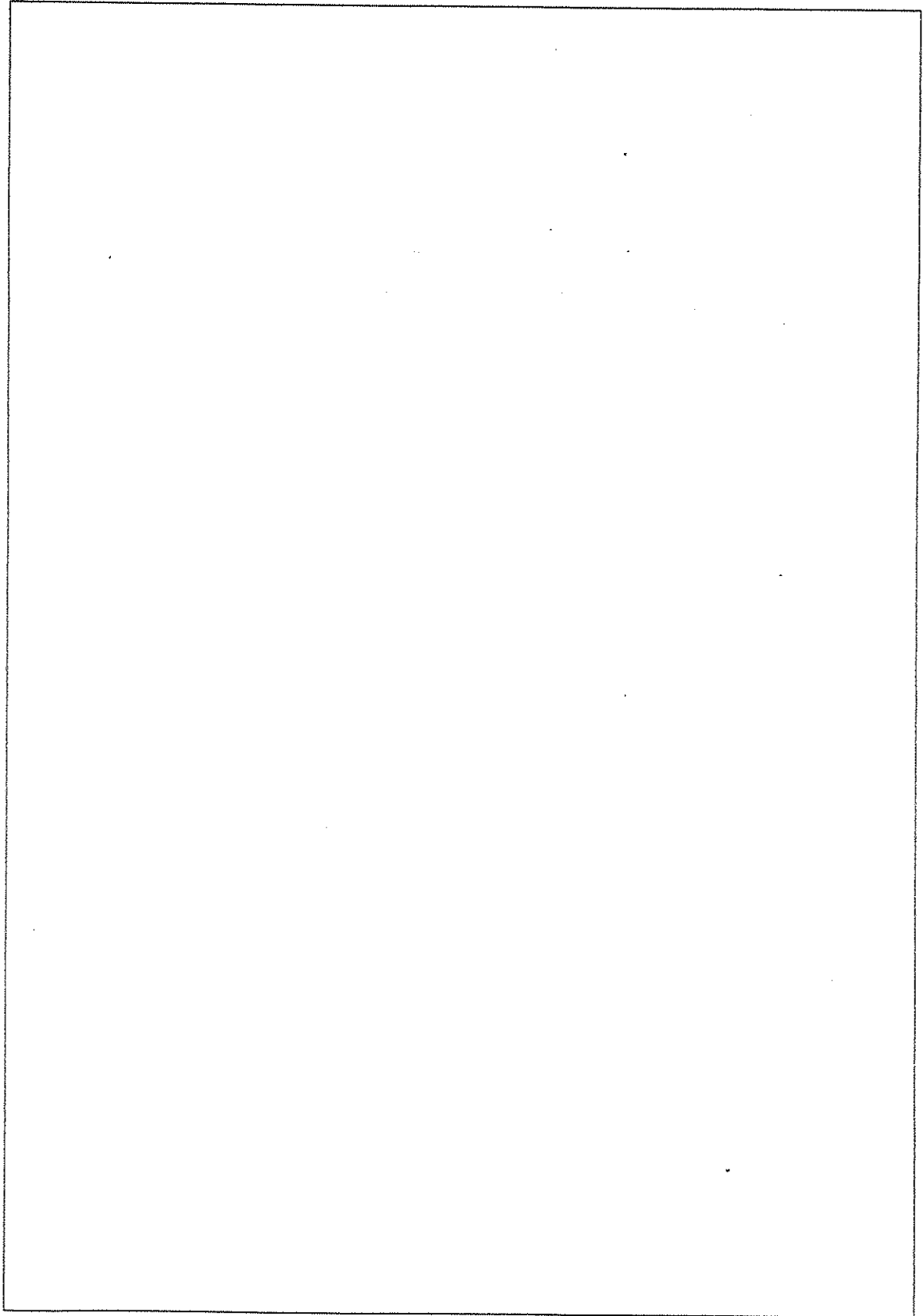
Index No:



(c) Briefly explain the stages in the Business Process Management life cycle. [9 marks]



Index No:



Index No:

Question 02 (30 Marks)

- a) Differentiate Virtual Machines vs. Containers (Give 2 factors).

[2 marks]

- b) DevOps was first intended to be a technique to organize engineering teams to effectively incorporate customer feedback. Elaborate briefly on this statement considering DevOps at the Organizational Level.

[8 marks]

Index No:

c) Consider the below code snippet of a design pattern implementation.

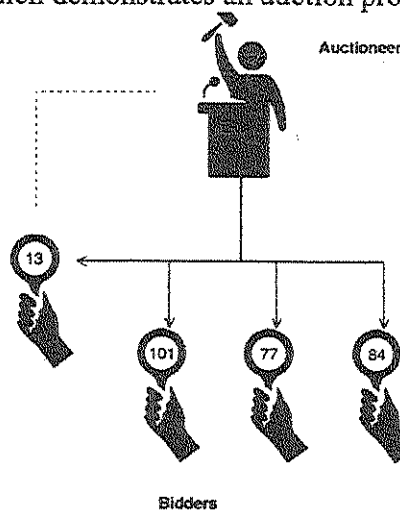
```
public class Examination {  
    private Examination() {}  
  
    private static class ExaminationHolder {  
        private static final Examination INSTANCE = new Examination();  
    }  
  
    public static Examination getInstance() {  
        return ExaminationHolder.INSTANCE;  
    }  
}
```

(i) What is the name of the design pattern depicted above? [1 mark]

(ii) What type of design pattern is it? [1 mark]

(iii) Briefly explain this pattern and why it is used. [2 marks]

d) Consider the below diagram which demonstrates an auction process.



Index No:

- (i) What is the name of the behavioural pattern corresponding to the above real-world analogy? [1 mark]

- (ii) What are the terms used for the special classes in the design patterns context that corresponds to the above scenario. [3 marks]

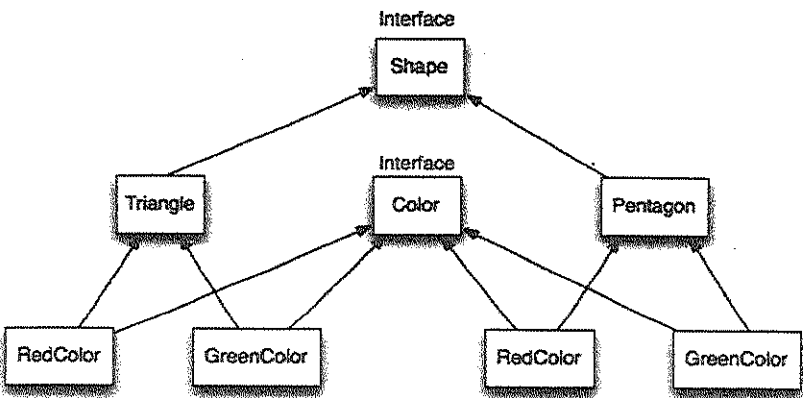
- (iii) Explain briefly about the design pattern you identified above. [2 marks]

- e) Consider Structural design patterns in Software Engineering.

- (i) Comment about the Open/Closed principle and Single Responsibility principle related to the Bridge design pattern. [2 marks]

- (ii) Let's imagine implementations and interfaces have an interface hierarchy similar to the one in the accompanying image below. We may separate the implementation from the interfaces by utilizing the bridge design pattern. After using the bridge pattern, draw the UML diagram for the classes and interfaces. [8 marks]

Index No:



Question 03 (25 Marks)

Following is a case study for a FlyMiles Air Line's flight reservation System.

Note: The quality tactic frameworks and general scenarios are given separately on the last page.

This system is used by various stakeholders, such as individuals, travel agents, and airline systems. Users can perform transactions related to making ticket reservations (which includes booking, reserving, and canceling tickets), checking the ticket status, and various other operations. Users can access the system using different modes e.g. by phone, by web, or at the information desk. In reserving a flight, the user first provides the details such as his personal information, desired travel dates, departure and destination cities and seat grade desired (first class, business class, or economy). Then the system informs the customer of all available flights with details that meet his or her criteria. If the customer finds any of the available flights acceptable, he or she may either pay for the ticket via credit card or request that the seat be held for 24 hours.

The customer has two ticketing options: they may request that a "paper" ticket be issued which is then posted to the home address, or an electronic ticket (E-ticket) may instead be assigned, in which case the customer is simply informed of the E-ticket serial number by SMS. In either case, the system records the serial number of the ticket issued to this customer.

It is essential that the system validates the input data and gives feedback to the user in case of errors or inconsistency. It is also essential that the system responds to customer queries in less than 10 milliseconds and should be able to handle at least 5,000 transactions/inquiries per second. The system should be available at any time.

- a) List four different types of stakeholders of this System

[2 marks]

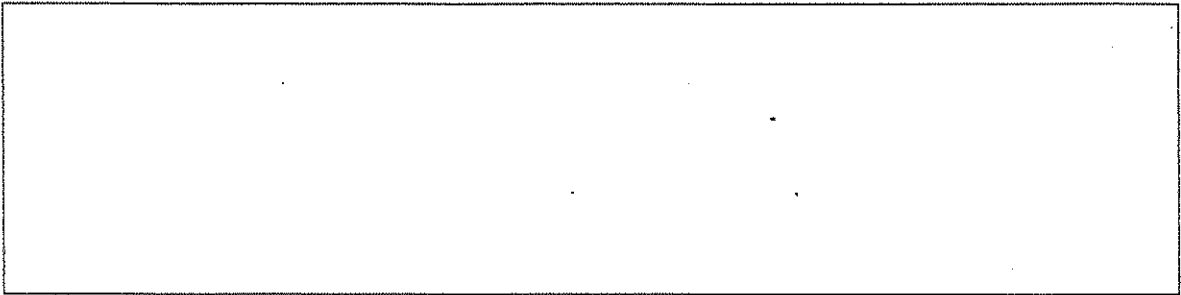
--

- b) Name two most important quality attributes for each of the four stakeholders.

[4 marks]

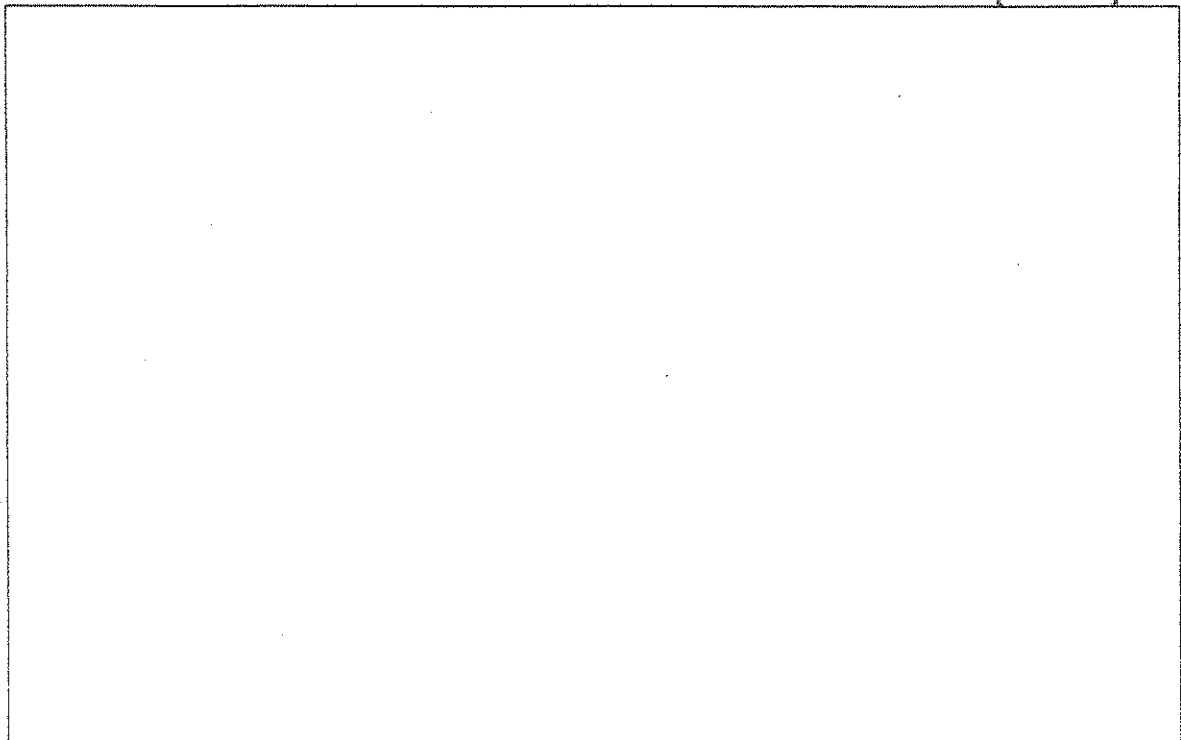
--

Index No:



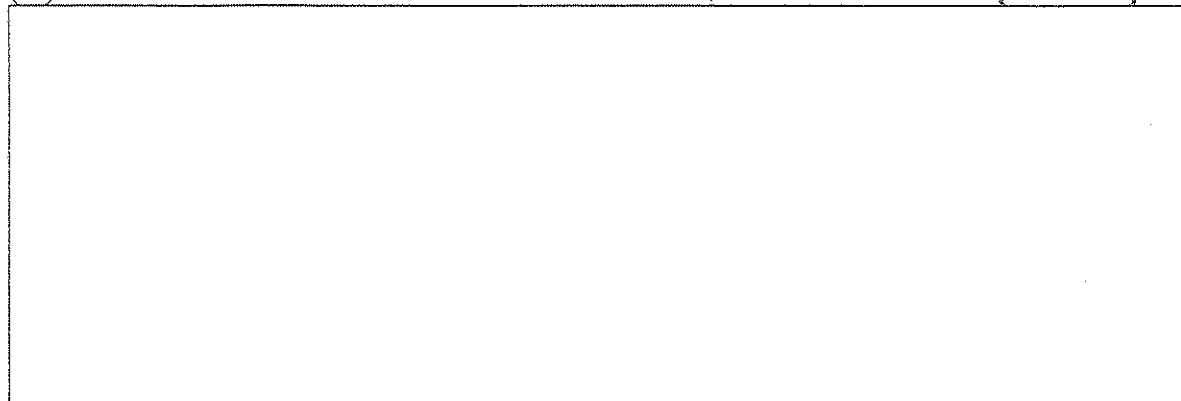
- c) Write one concrete quality attribute scenario for the performance of this System.

[5 marks]

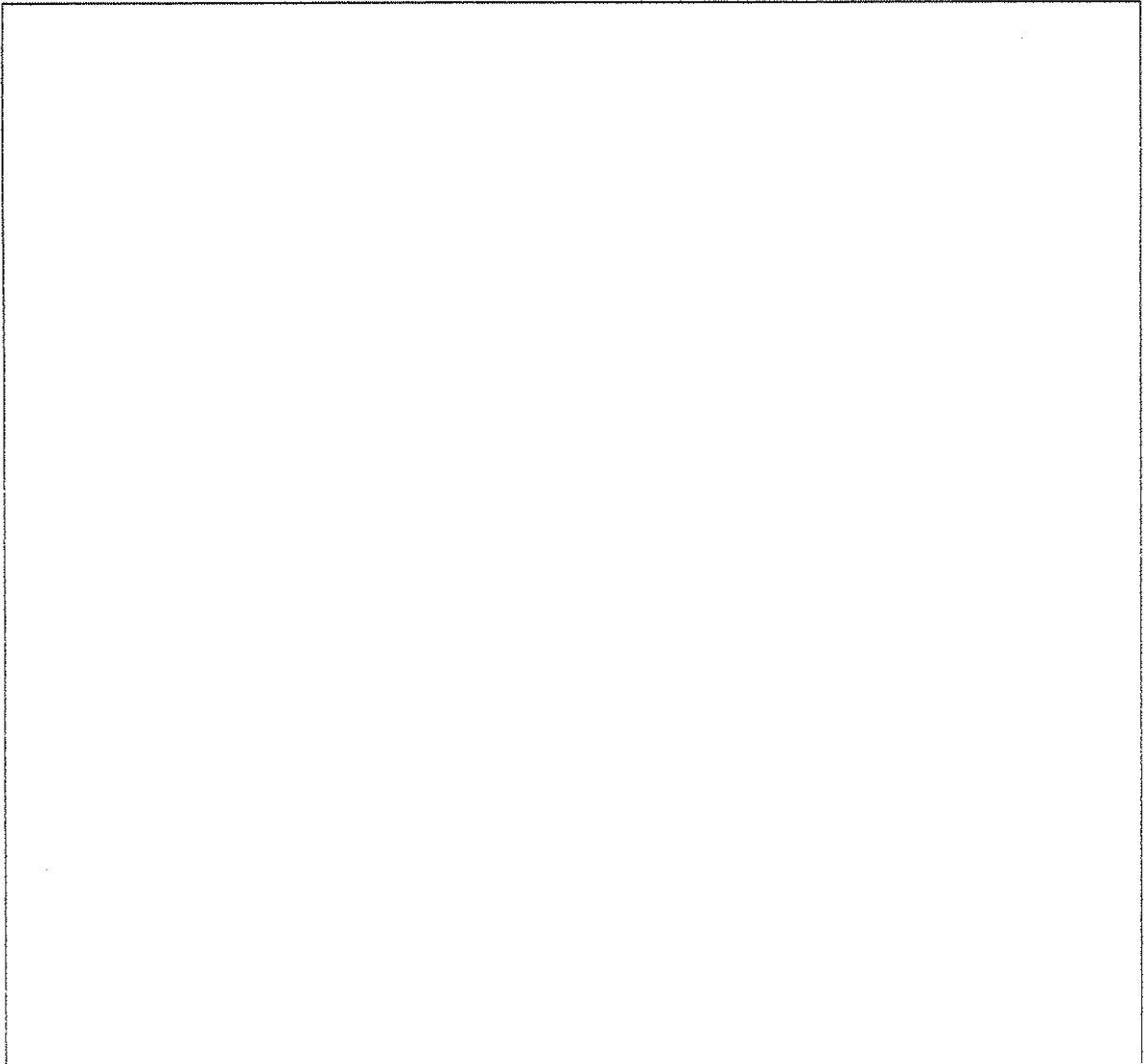


- d) Based on the performance tactics framework, explain the two most important tactics you are going to use in order to achieve the concrete quality attribute scenario you described in part (iii).

[10 marks]

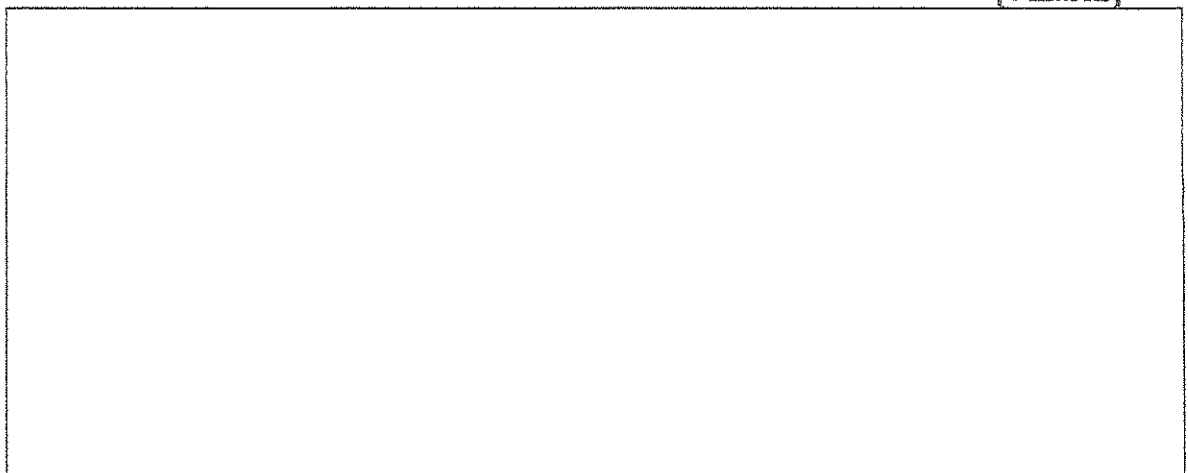


Index No:



e) Briefly explain four performance response measurements (one sentence each).

[4 marks]

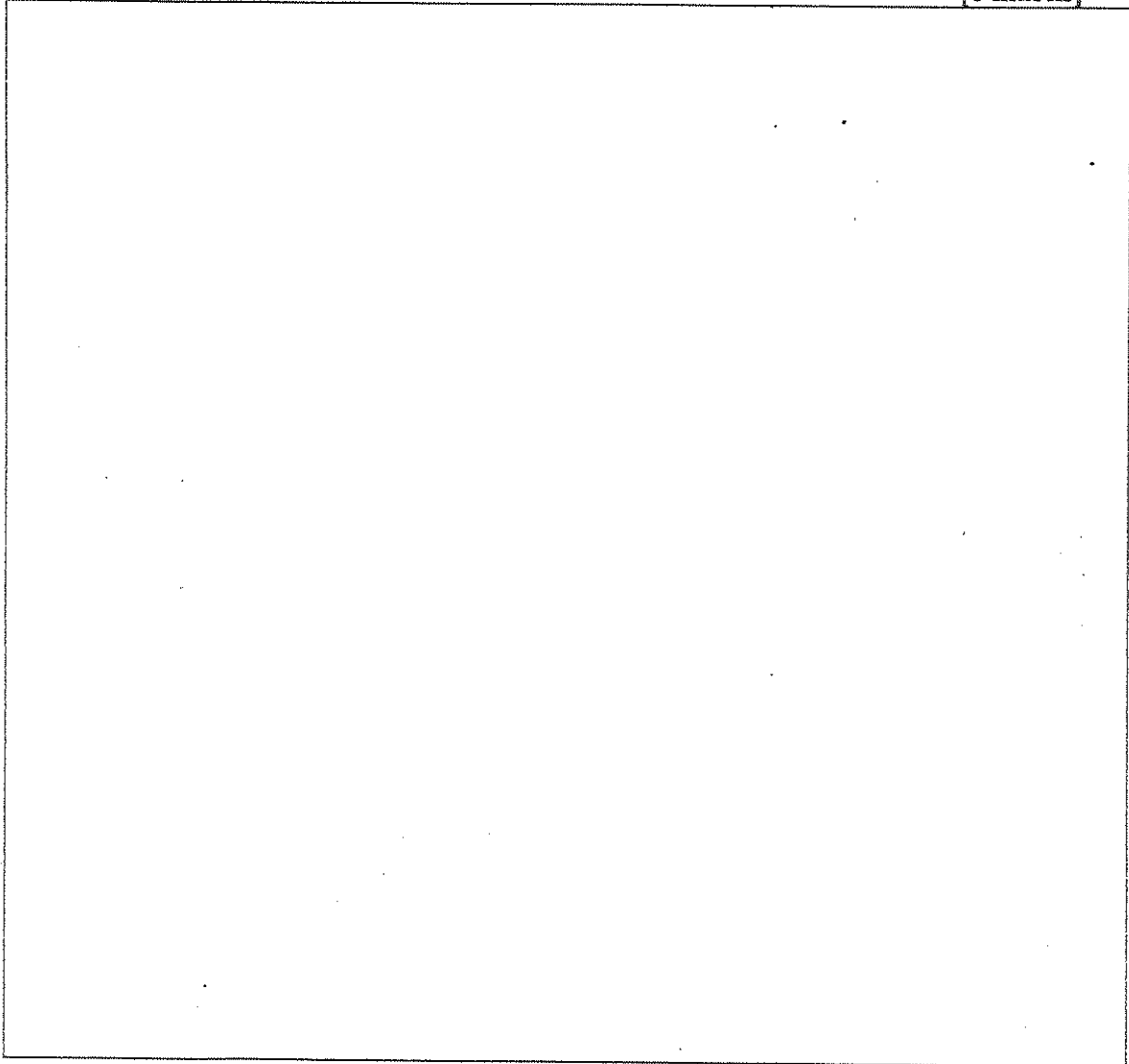


Index No:

Question 04 (25 Marks)

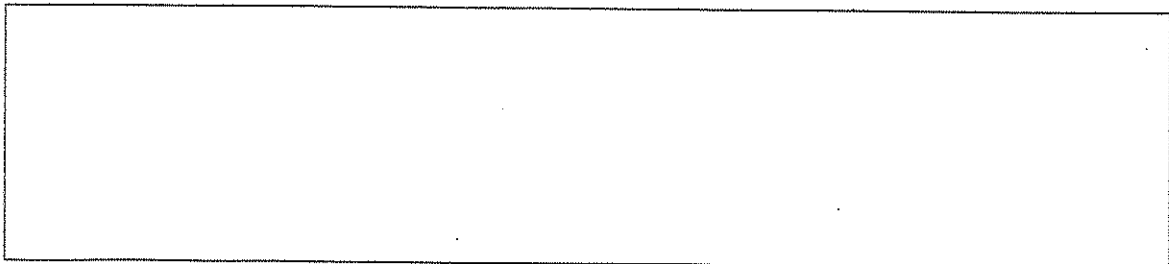
- a) Software Engineering projects can be categorized as Evolutionary Projects, Greenfield Projects and Framework-based Projects. Briefly explain these three (3) terms.

[6 marks]

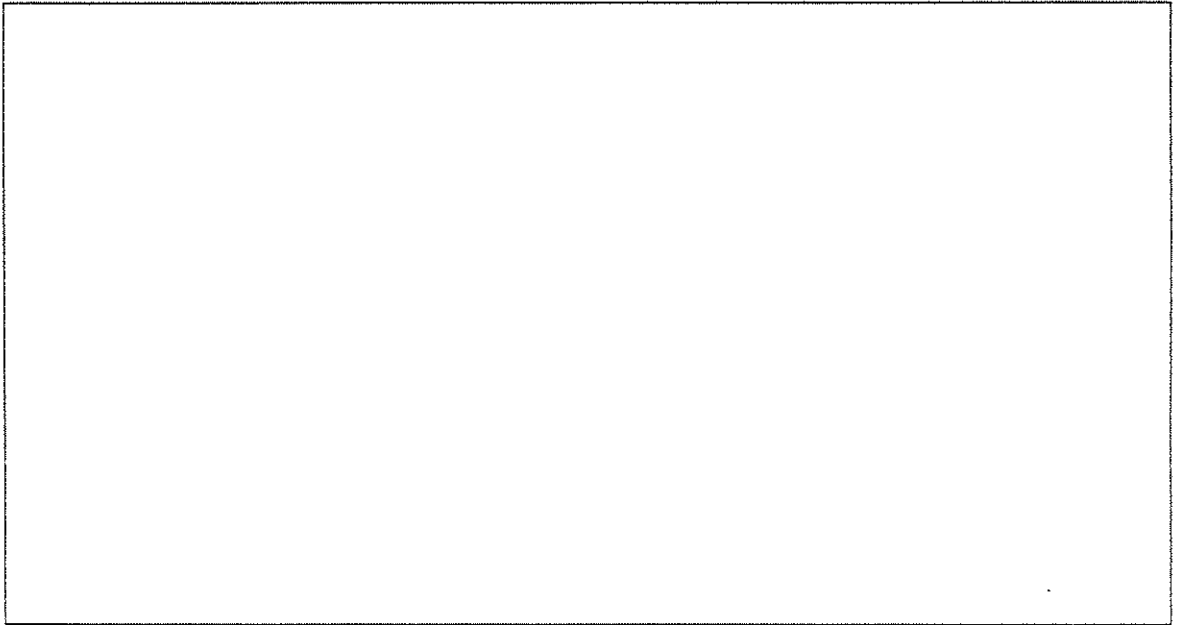


- b) "When an architect designs the architecture for a system, he/she has to consider about the quality attribute trade-offs." Justify this statement.

[5 marks]

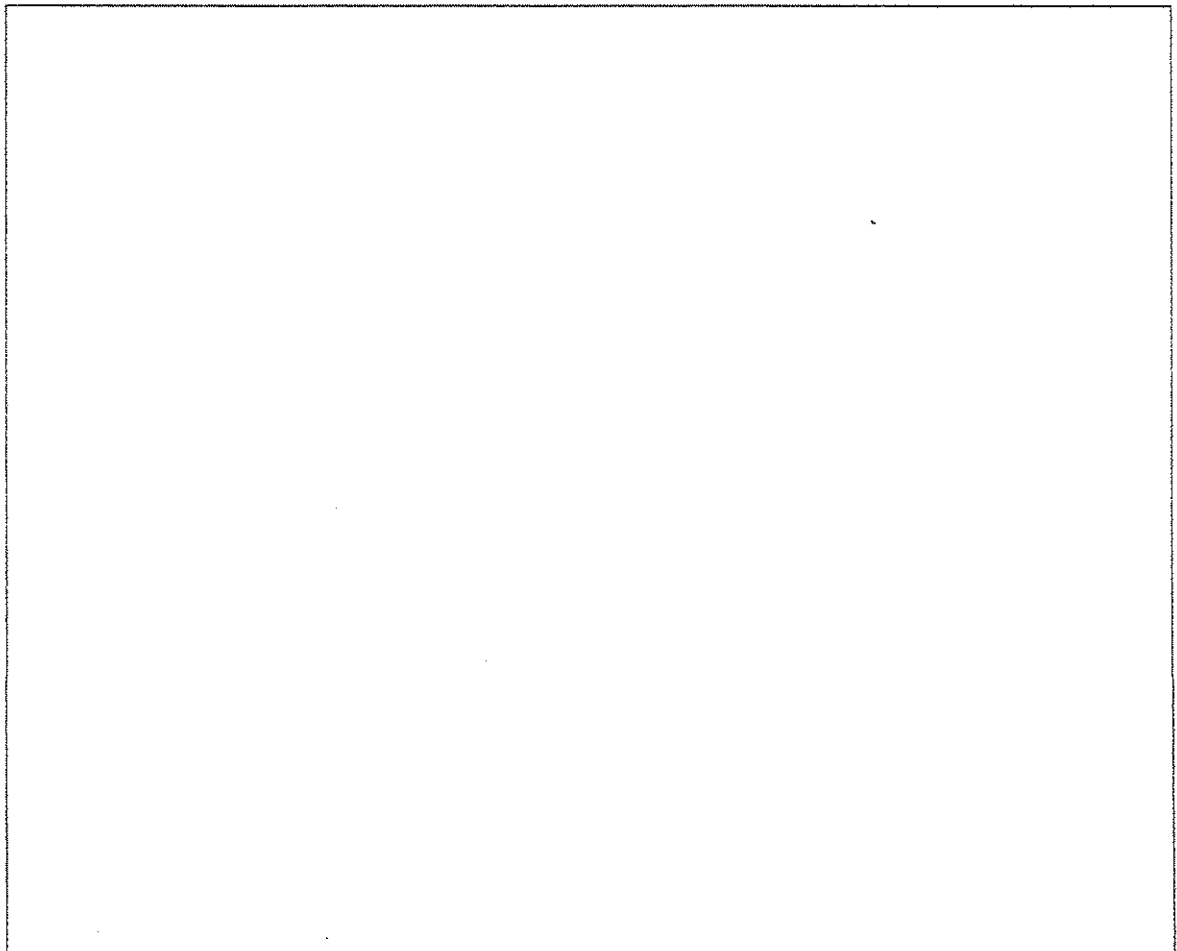


Index No:



c) Explain what an “Architectural Strategy” is.

[5 marks]



Index No:

- d) "Patterns implement multiple tactics". Explain this statement using an example.

[5 marks]

- e) Scheduling is a common tactic to support resource arbitration. List 4 commonly used scheduling policies.

[4 marks]

Index No:

Portion of Scenario	Possible Values
Source	One of a number of independent sources, possibly from within system
Stimulus	Periodic events arrive; sporadic events arrive; stochastic events arrive
Artifact	System
Environment	Normal mode; overload mode
Response	Processes stimuli; changes the level of service
Response Measure	Latency, deadline, throughput, jitter, miss rate, data loss

